CELEBRATING THE 30TH ANNIVERSARY OF TDTU (1997 - 2027)

UAV CONTEST 2027

The 4th International Competition on Research and Design of Unmanned Aerial Vehicles

"CONQUERING AUTONOMOUS LOW-ALTITUDE LOGISTICS — MULTI-AGENT CPS CHALLENGE"

From Single-Agent Waypoints to Decentralized Cyber-Physical Coordination via Open CABP Protocol

OFFICIAL MATCH DATES
SEPTEMBER 20 & 21, 2027

COMPETITION VENUE
TDTU SPORTS ARENA, TAN PHONG CAMPUS
Ho Chi Minh City, Vietnam

TRACK 01 — Single-UAV Class

STANDARD AUTONOMOUS TRACK

Evaluation of precision piloting, Edge-AI vision, and dynamic target delivery across 3 rigorous tournament stages:

- **Preliminary:** Gate transit + fixed D-point payload drop.
- **Semi-Final:** Randomized positions via real-time Computer Vision.
- **Final:** Dynamic rendezvous with **Mobile Receiver (10 cm/s)**.

Max Score: 290 pts — Time: 5 mins

INNOVATION BREAKTHROUGH

TRACK 02 • VOLUNTARY — Multi-Agent Swarm

COORDINATED OPS TRACK (2 UAVS)

Sequential relay pipeline simulating modern warehouse intralogistics: **"Place — Relay — Retrieve"**:

- **UAV-A (Manual):** Places package at candidate node ($T_1 \le 90\text{s}$).
- **UAV-B (Autonomous):** Autonomous retrieval ($T_2 \le 120\text{s}$).
- **Dual-Path Architecture:** Bonus Path (**CABP broadcast in 3s**) or Baseline Fallback via QR Vision.

Bonus: Up to +60 pts — Time: 7 mins

◊ Max Frame: ≤ 1.0 m^3 ◊ MTOW: ≤ 5.0 kg ◊ Battery: LiPo ≤ 6S (300 Wh) ◊ Lift Payload: ≥ 300 g ◊ Altitude: 3.0 - 5.0 m

</> **OFFICIAL OPEN-SOURCE STARTER KIT READY**
Complete reference implementation: Python UDP CABP protocol, state machines, and QR detection. Integrate in 15 mins!
github.com/CPAS-Lab

ARENA SPECIFICATIONS & COMPETITION GUIDELINES

Standardized $12 \times 24\text{ m}$ Indoor Testbed • Cyber-Physical Navigation Architecture

RULES V2.0 / APPROVED

ARENA LAYOUT SCHEMATIC ($12.0 \TIMES 24.0\TEXT{ M}$)

GPS-Denied Flight Environment



- Gate Center Height: 1.7 m (Peak 2.6m)
- D-Points QR Matrix: $15 \times 15\text{ cm}$
- Mobile Receiver Speed: 10 cm/s

ELIGIBILITY & ROSTER QUOTAS

Open to universities, research institutes, and engineering colleges worldwide:

- **Undergraduate Students:** Maximum 06 members
- **Graduate (Master's) Students:** Maximum 02 members
- **PhD Researchers:** Maximum 01 member
- **Total Team Roster:** ≤ 06 participants

Arena Flight Zone access: Max 02 members simultaneously.

AWARDS & RECOGNITION POOL

¤ GRAND CHAMPION TROPHY	Gold Medal & Cup
1st Runner-Up (Second Place)	Silver Medal
2nd Runner-Up (Third Place)	Bronze Medal
★ BEST INNOVATION AWARD	Multi-Agent Trophy

All participating teams receive official Certificates of Excellence.

REGISTRATION • IMPORTANT DATES

SCAN TO REGISTER & DOWNLOAD FULL RULEBOOK

Access complete technical specifications, CAD arena dimensions, starter source code SDK, and secure team registration portal.

Partner Tier	$5,000 USD (Co-Organizer, 1 Board Seat, 2 Teams)	Domestic: 2.5M VND	Intl: $350 USD ($500/2)

Registration Deadline	Code Checksum Lock	Tournament Kickoff
August 01, 2027	T - 30 Mins Pre-Match	Sept 20 & 21, 2027

OFFICIAL PORTAL

- Email: uavcontest2027@tdtu.edu.vn
- Portal: uavcontest.tdtu.edu.vn
- Tech Hotline: (+84) 918850014